# Foundation of Embodied AI: Course Map

Technical Tutorial

2026-10-20

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| Item | Details |
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| Topic | Course Map |
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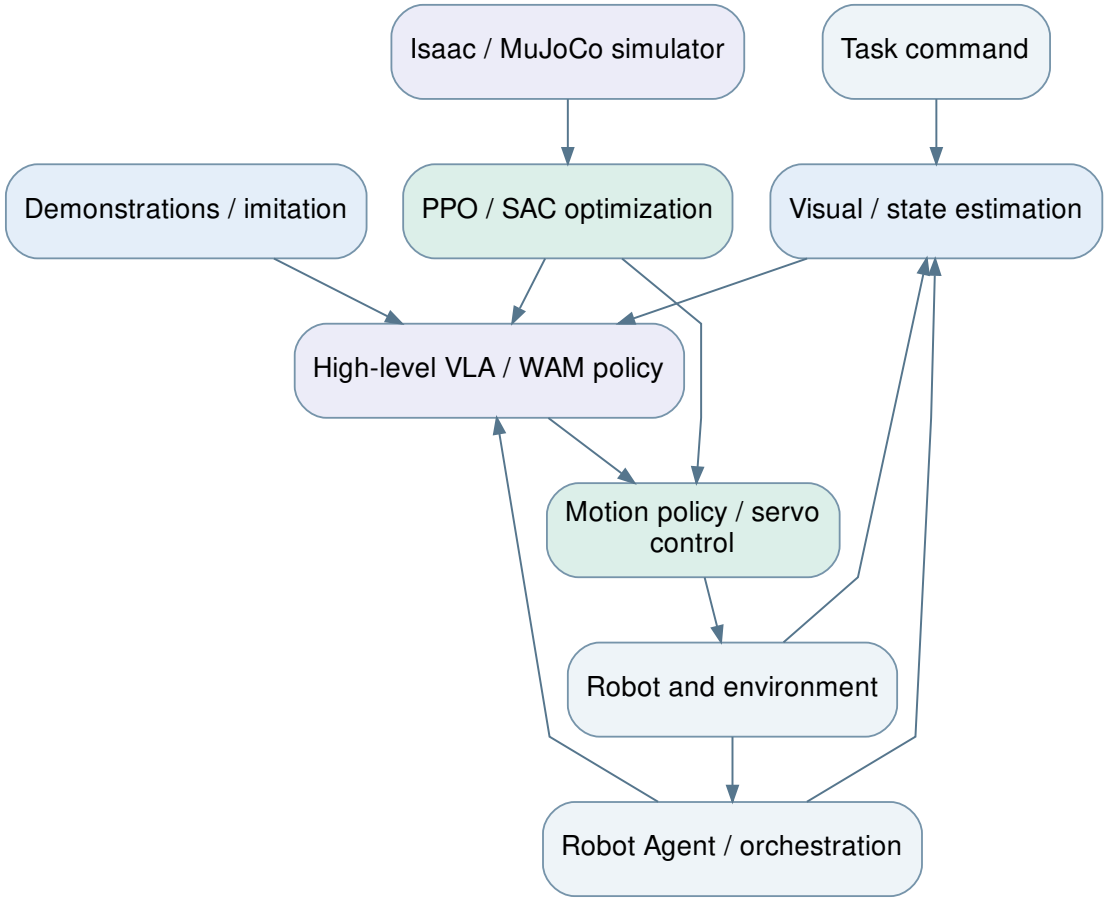
## Six learning modules

Seventeen topics are organized as tools → calibration and deep learning → physics and perception → robot RL → demonstrations and generative policies → systems integration. Long-form derivations remain optional deep dives rather than strict prerequisites.

| Module | Topics | Outcome |
|---|---|---|
| 0. Orientation | Systems map and guide | Explain perception–policy–control loops |
| 1. Tools | Git, ROS 2 | Manage code and understand ROS communication and TF |
| 2. Physics, perception and neural foundations | Calibration, Deep Learning Basics, vision foundation models, dynamics, Isaac / MuJoCo | Relate calibrated geometry, neural networks, physical models and simulation |
| 3. Reinforcement learning | Motion RL, PPO, SAC | Design observations, actions, rewards and data collection |
| 4. Demonstrations and models | Imitation Learning, Flow Matching, GR00T / SONIC, World Action Models, Efficient Training & Inference, Post-training | Learn generative and predictive policies and connect efficient inference with hierarchical controllers |
| 5. Robot Agent integration | Robot Agent planning, tool use, verification and recovery | Orchestrate perception, policy and control for long-horizon tasks |

## Embodied AI system map




## Reading paths

1. Beginners: orientation → essential Git and ROS 2 → calibration → deep learning → dynamics → vision foundation models → simulation.
2. Motion control: dynamics → simulation → deep learning → PPO / SAC → motion control RL.
3. VLA: calibration → deep learning → vision foundation models → Imitation Learning → Flow Matching → GR00T / SONIC → World Action Models → efficient training and inference → post-training → Robot Agent.
4. Mathematical deep dives: full PPO, SAC, and Flow Matching derivations are reference material; hands-on learning does not require deriving everything first.

## Tutorial overview

### 1. Git Basics

Git Basics — Learn Git's working tree, staging area and commit history, then follow a GitHub workflow covering branches, pull requests, conflict resolution and common recovery commands.

### 2. ROS 2 Basics

ROS 2 Basics — Understand nodes, topics, services, actions, QoS and TF. Connect sensors, state estimation, policy inference and controllers through ROS 2 interfaces.

### 3. Camera Calibration

Camera Calibration — Study camera intrinsics, distortion, extrinsics and hand–eye calibration. Relate RGB-D observations and coordinate transforms to robot-frame object poses.

### 4. Deep Learning Basics

Deep Learning Basics — Learn supervised objectives, gradient descent and backpropagation, then connect MLPs, Attention and Transformers to multimodal robot action heads. Understand pretraining, fine-tuning and the foundations required by downstream policy methods.

### 5. Vision Foundation Models

Vision Foundation Models — Compare segmentation, pose estimation, stereo depth and 3D foundation models. Follow four robotics workflows spanning known objects, stereo geometry, unknown shapes and human demonstrations.

### 6. Robot Dynamics

Robot Dynamics — Develop rigid-body motion and dynamics through coordinate frames, Jacobians, inertia and equations of motion. Connect contact, friction and control models to robot simulation.

### 7. Isaac / MuJoCo Simulation

Isaac / MuJoCo Simulation — Understand Isaac Sim, Isaac Lab and MuJoCo as simulation and training tools. Cover asset import, environment contracts, RL library selection, parallel throughput, deployment interfaces.

### 8. Motion Control RL

Motion Control RL — Study motion tracking, DeepMimic/AMP, universal controllers, teacher–student learning, RMA and residual RL. Connect these methods to legged and dexterous motion, recovery and Sim-to-Real.

### 9. PPO / Actor–Critic

PPO / Actor–Critic — Derive PPO-Clip and Actor–Critic learning from policy gradients, Bellman targets and GAE. Explain Gaussian policy updates, gradient isolation and asymmetric robot observations.

### 10. Q-learning / SAC

Q-learning / SAC — Trace Q-learning, TD/SARSA and DQN toward entropy-regularized RL. Derive soft Bellman equations and modern SAC with twin critics, reparameterization and automatic entropy tuning.

### 11. Imitation Learning

Imitation Learning — Introduce BC, DAgger and ACT, then connect Diffusion Policy and Flow Matching to action-chunk generation. Explain temporal ensembling and inference-time versus training-time RTC.

### 12. Flow Matching

Flow Matching — Derive conditional Flow Matching, velocity fields and the training objective. Explain how ODE integration transforms noise into conditional robot action sequences.

### 13. GR00T & SONIC

GR00T & SONIC — Use GR00T and SONIC to explain System 1/System 0 humanoid control. Connect high-level action chunks, motion representations, whole-body policies and low-level execution.

### 14. World Action Models

World Action Models — Distinguish world models, VLAs and WAMs, then study DreamZero training and accelerated inference, Dream-1 cross-embodiment deployment, and VLA/WAM memory, compute and latency trade-offs. Connect these methods to contact-rich control, recovery and evaluation.

### 15. Efficient Training & Inference

Efficient Training & Inference — Connect Transformer and VLA/WAM training costs, FlashAttention and valid caching to DreamZero acceleration, few-step inference, asynchronous robot control, RTC and cloud latency.

### 16. Post-training

Post-training — Compare SFT and RL for pretrained VLA and WAM models. Study six VLA RL routes, WAM-RL with online video SFT and Flow-SDE, then connect actor–critic semantics, RLinf integration and robot deployment limits.

### 17. Robot Agents / RPent

Robot Agents / RPent — Distinguish foundation models, coding agents and agent harnesses. Connect RAX exploration, ASPIRE trace repair and RPent toolkits, memory, Flash Mode and the RLinf data flywheel to reusable robot workflows.